Problem Set 5

Question 1 : pp 439 - 440

Real exchange rate, $\mathcal{E} = \frac{EP}{P^*}$

And hence % change in real exchange rate,

$$\% \Delta \text{ in } \mathcal{E} = \% \Delta \text{ in } E + \% \Delta \text{ in } P - \% \Delta \text{ in } P^*$$

When nominal exchange rate is fixed, $\% \Delta \text{ in } E = 0$

$$\% \Delta \text{ in } \mathcal{E} = \% \Delta \text{ in } P - \% \Delta \text{ in } P^*$$

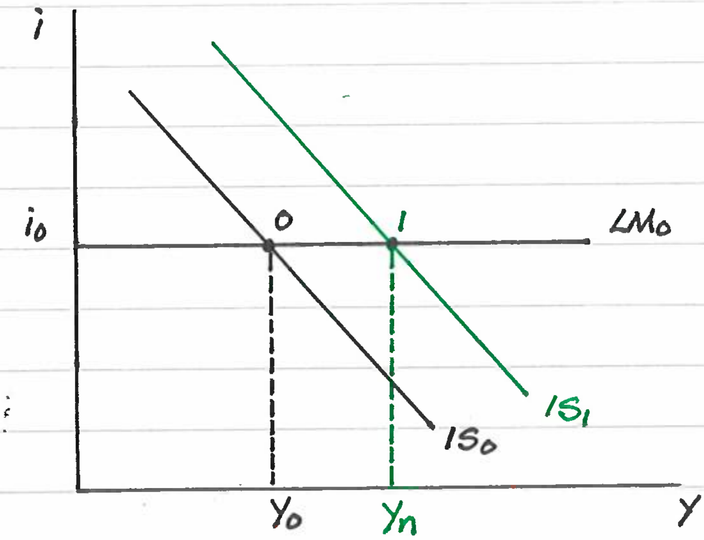
$$\boxed{\% \Delta \text{ in } \mathcal{E} = \pi - \pi^*}$$

(a) If foreign economy is in medium run eq^{m}, then $Y^* = Y_n^* \Rightarrow \pi^* = \bar{\pi}^*$

(b) Suppose $\bar{\pi} = \bar{\pi}^*$, then

$$\left.\begin{array}{l}\pi - \bar{\pi} = (\frac{\alpha}{L})(Y - Y_n) \Rightarrow Y = Y_n \text{ and } \pi = \bar{\pi} \\ \pi^* - \bar{\pi}^* = (\frac{\alpha^*}{L^*})(Y^* - Y_n^*) \Rightarrow Y^* = Y_n^* \text{ \& } \pi^* = \bar{\pi}^*\end{array}\right\} \pi = \pi^* = \bar{\pi} = \bar{\pi}^*$$

(c) When domestic currency is overvalued, domestic goods are made relatively expensive, the NX is low. Then, $NX < 0$ and $Y < Y^*$.



Initially $IS = IS_0$ and $LM = LM_0$. Eq^{m} at pt 0. $Y = Y_0 < Y_n$ and $i = i_0$ and $NX_0 < 0$

As $Y < Y_n$ → Recessionary gap → In the medium run, $P \downarrow$ and $\mathcal{E} = \frac{\bar{E}P}{P^*} \downarrow$

Real dep → X cheaper → X ↑ → NX ↑ → $\vec{IS}$ ($IS_0 \to IS_1$)
→ M expensive → M ↓

(d) Same as (c). Immediate devaluation → real dep → $\varepsilon \downarrow$ → $NX \uparrow$ → $\vec{IS}$ and the economy returns to Y_n immediately rather than gradually through $P \downarrow$

(e) Under fixed exchange rate, $i = i^*$.
In the period of devaluation, the return on domestic bonds is lower than the return on foreign bonds.
It is hard to know if the bondholders will believe the devaluation is one-time only. If they believe another devaluation is possible, then it will be higher than foreign interest rate.

Question 2

The UIP suggests that

$$i_t = i_t^* - \underbrace{\frac{E_{t+1}^e - E_t}{E_t}}_{\text{Expected dep rate of foreign currency}}$$

If $i_t = 5\%$ and $i_t^* = 4\%$, for financial investors to hold both domestic and foreign bonds, $\frac{E_{t+1}^e - E_t}{E_t} = -1\%$ → Exp. dep rate of foreign currency is -1%

Exp. app rate of foreign currency is 1%

(a) If $i_t^* = 3$, $E_t = 0.5$ and $E_{t+1}^e = 0.5$

Expected dep rate of foreign currency, $\frac{E_{t+1}^e - E_t}{E_t} = 0$

Then $i_t = 3$

(b) In period 2, when crisis begins, $i_t^* = 3$, $E_t = 0.5$ but $E_{t+1}^e = 0.45$

Expected dep rate of foreign currency, $\frac{E_{t+1}^e - E_t}{E_t} = -10\%$

Expected dep rate of domestic currency is 10%

$$i_t = 0.03 - \left(\frac{0.45 - 0.5}{0.5}\right)$$

$$i_t = 0.13 \text{ or } 13\%$$

(c) The resolution of the crisis in period 4 occurs when beliefs change about the expected future exchange rate i.e. $E_{t+1}^e = 0.5$ again. The peg becomes credible.

(d) In period 5, $\frac{E_{t+1}^e - E_t}{E_t} = \frac{0.4 - 0.5}{0.5} = -20\%$

Expected dep rate of domestic currency is 20%

$$i_t = 0.03 - \left(\frac{0.4 - 0.5}{0.5}\right)$$

$$i_t = 0.23 \text{ or } 23\%$$

The domestic interest rate has to rise to 23% to maintain UIP. It only rises to 15 which is not sufficient. To prevent dep, central bank has to buy domestic currency and sell foreign reserves. Foreign reserves at the central bank will decrease.

(d) Since $E_t = E_{t+1}^e = 0.4$ instead of 0.5, this means that a devaluation has occurred. It may be difficult for the govn / central bank to convince bondholders that they will never devalue again.